



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 11

Subject:	DevOps Lab
TCODE:	TE7950
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Branch	CSE, Batch (2023-27), L4
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	23 rd Spte, 2026
Title of Assignment:	Docker Compose for Multi-Container App
Theory:	Docker Compose is a tool used to define and manage multi-container Docker applications using a single docker-compose.yml file. It allows multiple services, such as a web application and a database, to be configured, started, and managed together with a single command. Docker Compose automatically creates a network for communication between services, manages named volumes for persistent storage, and simplifies deployment by eliminating the need to start each container individually. This makes it easier to develop, test, and maintain applications that consist of multiple interconnected containers. Commands Used • Created a new project directory and navigated into it: mkdir my-multi-app cd my-multi-app • Started the multi-container application in detached mode: docker compose up -d • Verified the running containers: docker compose ps (or <i>docker ps</i> if <i>docker compose ps</i> was not available) • Stopped and removed the containers and network created by Docker Compose: docker compose down

Step 1: Create the Project Directory

Created Project Directory — Created a new project directory named my-multi-app and navigated into it to store the Docker Compose configuration.

Step 2: Create docker-compose.yml

Created Docker Compose Configuration — Created and saved the docker-compose.yml file defining the Flask web service, Redis service, Docker volumes, and networking configuration for the multi-container application.

Step 3: Start the Multi-Container Application

Started Multi-Container Application — Started the multi-container application using Docker Compose, which created the required network, volumes, and launched both the Flask web service and Redis service.

Step 4: Access the Web Application

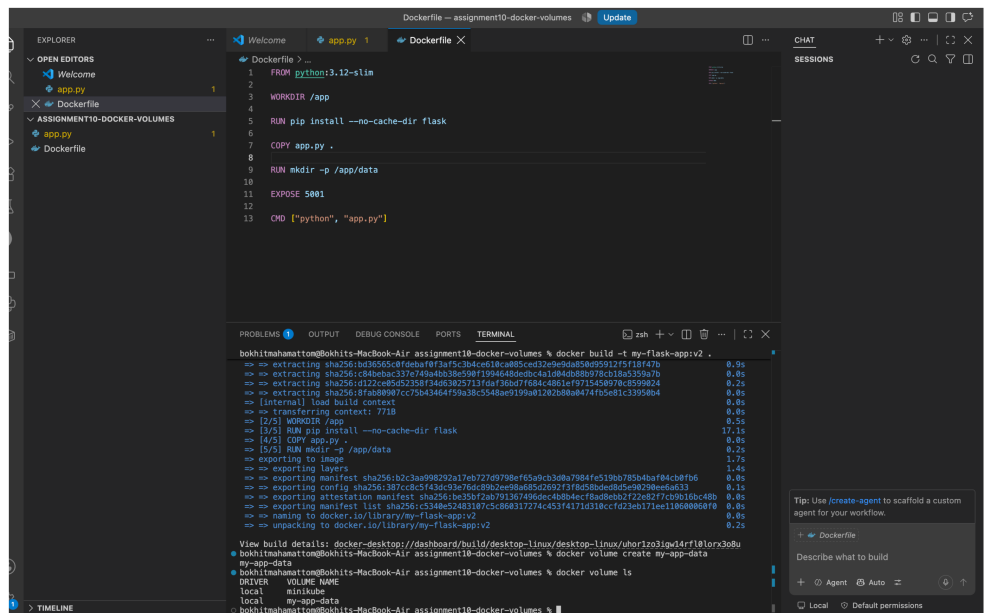
Verified Web Application — Accessed the Flask application through <http://localhost:8083> and verified that the web service was successfully running as part of the Docker Compose application.

Step 5: Verify the Running Services

Verified Running Services — Verified that the Docker Compose application was running successfully by listing the active containers, including the Flask web service and the Redis service.

Step 6: Stop and Remove the Multi-Container Application

Stopped Multi-Container Application — Stopped and removed the Docker Compose application, including the running containers and the automatically created network, using the docker compose down command.



The screenshot displays the Docker Desktop interface. On the left, the Explorer pane shows the project structure with files like 'app.py', 'Dockerfile', and 'ASSIGNMENT10-DOCKER-VOLUMES'. The main editor shows the 'Dockerfile' with the following content:

```
1 FROM python:3.12-slim
2
3 WORKDIR /app
4
5 RUN pip install --no-cache-dir flask
6
7 COPY app.py .
8
9 RUN mkdir -p /app/data
10
11 EXPOSE 5001
12
13 CMD ["python", "app.py"]
```

The bottom pane shows the terminal output of the command `docker build -t my-flask-app:v2 .`. The output details the build process, including extracting the Python base image, installing Flask, copying the application code, and creating the final image. The build is successful, resulting in a 0.2s duration.

Below the terminal output, the 'View build details' section shows the build configuration for the 'my-app-data' volume, including the driver (local), volume name (minikube), and the command to create the volume.

Hello, Docker!

Persistent Logs:

2026-10-01 06:12:36: Hello, Docker!

Hello, Docker Compose!

Flask + Redis

Visit count: 5

Time: 2026-10-01 07:44:29

Hello, Docker Compose!

Flask + Redis

Visit count: 12

Time: 2026-10-01 07:45:53

Hello, Docker Compose!

Flask + Redis

Visit count: 13

Time: 2026-10-01 07:54:34

```

bokhitmahamattom@Bokhits-MacBook-Air assignment11-docker-compose % docker compose up -d --build
=> resolving provenance for metadata file 0.0s
[+] up 6/6
✓ Image assignment11-docker-compose-web Built 9.7s
✓ Network assignment11-docker-compose_default Created 0.0s
✓ Volume assignment11-docker-compose_redis-data Created 0.0s
✓ Volume assignment11-docker-compose_app-data Created 0.0s
✓ Container redis-compose-app Started 0.4s
✓ Container flask-compose-app Started 0.4s
● bokhitmahamattom@Bokhits-MacBook-Air assignment11-docker-compose % docker compose ps
NAME IMAGE COMMAND SERVICE CREATED
ATUS PORTS
flask-compose-app assignment11-docker-compose-web "python app.py" web About a minute ago
About a minute 0.0.0.0:8083->5001/tcp, [::]:8083->5001/tcp
redis-compose-app redis:7-alpine "docker-entrypoint.s..." redis About a minute ago
About a minute 6379/tcp
● bokhitmahamattom@Bokhits-MacBook-Air assignment11-docker-compose % docker compose down
[+] down 3/3
✓ Container flask-compose-app Removed 3.2s
✓ Container redis-compose-app Removed 0.2s
✓ Network assignment11-docker-compose_default Removed 0.1s
○ bokhitmahamattom@Bokhits-MacBook-Air assignment11-docker-compose %

```

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